EXPERIMENT

Deterministic Experiment | **Random Experiment**



(Genetic determination)



(Hitting the target)

Definition 12.1

An **experiment** is defined as a process for which its result is well defined.

Definition 12.2

Deterministic experiment is an experiment whose outcomes can be predicted with certain, under ideal conditions.

Definition 12.3

A random experiment (or non-deterministic) is an experiment

(i) whose all possible outcomes are known in advance,

(ii) whose each outcome is not possible to predict in advance, and

(iii) can be repeated under identical conditions.

A die is 'rolled', a fair coin is 'tossed' are examples for random experiments.

Definition 12.4

A **simple event** (or elementary event or sample point) is the most basic possible outcome of a random experiment and it cannot be decomposed further.

Definition 12.5

A sample space is the set of all possible outcomes of a random experiment. Each point in sample space is an elementary event.

Illustration 12.1

(1) (i) If a die is rolled, then the sample space $S = \{1, 2, 3, 4, 5, 6\}$.

(ii) A coin is tossed, then the sample space $S = \{H, T\}$.

(2) (i) Suppose we toss a coin until a head is obtained. One cannot say in advance how many tosses will be required, and so the sample space.

$S = \{H, TH, TTH, TTTH, ...\}$ is an infinite set.

(ii) The sample space associated with the number of passengers waiting to buy train tickets in counters is $S = \{0,1,2,...\}$.

(3) (i) If the experiment consists of choosing a number randomly between 0 and 1, then the sample space is $S = \{ x: 0 < x < 1\}$.

(ii) The sample space for the life length (t in hours) of a tube light is $S = \{ t: 0 < t < 1000\}$.

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